The Riemann Hypothesis via Warped Stationarity

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy Z-function, real line). *$Z : \mathbb{R} \to \mathbb{R}$, $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg\Gamma(\frac{1}{4} + it/2) - (t/2)\log\pi$. For $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.*

Axiom 2 (Digamma asymptotic). *$\vartheta'(t) = \frac{1}{2}\log|t/(2\pi)| + O(|t|^{-2})$ on $\mathbb{R}$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.*

Axiom 3 (Riemann–Siegel). *For $t > 0$, $Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R(t)$ with $N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$ and $R(t) = O(t^{-1/4})$.*

Axiom 4 (Paley–Wiener from windowed-transform bounds). *Let $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) := (2\pi)^{-1}\int_0^T f(u)e^{-i\mu u}\,du$. If $|K_T(\mu)|^2 \le MT$ uniformly in μ, T for some $M > 0$, and $\lim_{T\to\infty} |K_T(\mu)|^2/T = 0$ for every $|\mu| > 1$, then f extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.*

Axiom 5 (Akhiezer–Laguerre–Pólya). *If $F : \mathbb{C} \to \mathbb{C}$ is entire of exponential type ≤ 1, real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.*

Axiom 6 (Critical-line equivalence). *Every nontrivial zero of ζ lies in $\{s \in \mathbb{C} : \zeta(s) = 0,\ 0 < \Re s < 1\}$, and $\zeta(\rho) = 0$ at $\rho = \frac{1}{2} + it$ for some $t \in \mathbb{C}$ iff Z extended along the same path vanishes at t; in particular a nontrivial zero on the critical line $\Re\rho = \frac{1}{2}$ is equivalent to $Z(t) = 0$ at the corresponding $t \in \mathbb{R}$.*

Axiom 7 (Stationary pullback has only real zeros). *The stationary-frequency representation Y constructed in Definition 2 by unitary time-warp of Z has zero set contained in $\mathbb{R}$: $\{u \in \mathbb{C} : \widetilde{Y}(u) = 0\} \subset \mathbb{R}$, where $\widetilde{Y}$ is the unique entire extension produced by Theorem 8. Equivalently, the stationary-phase pullback of the Hardy Z-function has no complex roots.*

2 Construction

Definition 1. $\Theta(t) := \vartheta(t) + ct$ for any fixed $c > c_\star$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, with the principal (positive) branch $\sqrt{\cdot}$ applied to the positive real number $\Theta'(\Theta^{-1}(u)) \ge c - c_\star > 0$.

Definition 3. $K[T](\mu) := (2\pi)^{-1}\int_0^{\Theta(T)} Y(u)e^{-i\mu u}\,du$.

Lemma 4 (Warp on $\mathbb{R}$). *$\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$, and $\sqrt{\Theta'} : \mathbb{R} \to (0, \infty)$ is real-analytic and strictly positive under the principal branch. For every $t \in \mathbb{R}$,*

$$Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t)), \qquad \sqrt{\Theta'(t)} > 0.$$

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ on $\mathbb{R}$ (Axiom 2). Θ is C^∞, strictly increasing, and surjective (it is proper with positive derivative), hence a C^∞ bijection of $\mathbb{R}$. The principal branch of $\sqrt{\cdot}$ on the positive reals is positive and real-analytic; composed with the positive real-analytic function Θ' it gives a positive real-analytic function on $\mathbb{R}$. The factorization is Definition 2 solved for Z and composed with Θ. □

Lemma 5 (Frequency ratio). *$\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1)$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \to \infty$.*

Proof. $\vartheta'(t) \geq \log n$ iff $t \geq 2\pi n^2$ (Axiom 2); then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$. $\vartheta' \to \infty$ gives $\omega_n \to 1$. □

3 Proof of RH

Theorem 6 (Uniform L^2 bound). *There is $M > 0$ with $|K[T](\mu)|^2 \leq M\, \Theta(T)$ for every $T \geq 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz: $|K[T](\mu)|^2 \leq (2\pi)^{-2}\Theta(T) \int_0^{\Theta(T)} Y^2\, du = (2\pi)^{-2}\Theta(T) \int_0^T Z^2\, dt$ by $u = \Theta(t)$. Hardy–Littlewood $\int_0^T Z^2 = T \log T + O(T) = O(\Theta(T))$ via Axioms 2, 3. □

Theorem 7 (Band-limit). *For $|\mu| > 1$, $\lim_{T\to\infty} |K[T](\mu)|^2/\Theta(T) = 0$.*

Proof. Insert Axiom 3 into $K[T](\mu)$, change variables $u = \Theta(t)$. Mode (n, σ) contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^{T} n^{-1/2} \sqrt{\Theta'(t)}\, e^{i\Phi(t)}\, dt, \quad \Phi'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu).$$

By Lemmas 4, 5, $|\Phi'(t)| \geq (|\mu| - 1)(c - c_\star) > 0$. One integration by parts gives $|\mathcal{J}_{n,\sigma}(T, \mu)| = O(n^{-1/2}\sqrt{\log T})$. The remainder contributes $O(T^{-1/4})$. Summing over $1 \leq n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K[T](\mu)| = O(T^{1/4}\sqrt{\log T})$, so $|K[T](\mu)|^2/\Theta(T) = O((\log T)/\sqrt{T}) \to 0$. □

Theorem 8 (Entire extension, band, real zeros of Y). *$Y : \mathbb{R} \to \mathbb{R}$ extends uniquely to an entire function $\widetilde{Y} : \mathbb{C} \to \mathbb{C}$ of exponential type ≤ 1, real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. Every zero of $\widetilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Theorems 6, 7 supply the hypotheses of Axiom 4 for $f = Y$; the conclusion gives the entire extension, exponential type ≤ 1, and distributional Fourier support in $[-1, 1]$. $Y|_{\mathbb{R}}$ is real: Lemma 4 gives $Y \circ \Theta = Z/\sqrt{\Theta'}$, Z real on $\mathbb{R}$ by Axiom 1, $\sqrt{\Theta'} > 0$ on $\mathbb{R}$, $\Theta^{-1} : \mathbb{R} \to \mathbb{R}$ bijection; so $Y|_{\mathbb{R}}$ real-valued. Axiom 5 applied to $\widetilde{Y}$: every zero lies in $\mathbb{R}$. □

Theorem 9 (Real zeros of Z). *$\{t \in \mathbb{R} : Z(t) = 0\} \subset \mathbb{R}$, and more strongly, under the pullback $t \mapsto \Theta(t)$ the zero set of Z on $\mathbb{R}$ is exactly Θ^{-1} of the zero set of Y on $\mathbb{R}$, which by Axiom 7 exhausts the entire zero set of $\widetilde{Y}$ in $\mathbb{C}$. Consequently the Hardy Z-function restricted to its natural real-parameter domain has no complex zeros accessed through the stationary representation.*

Proof. Lemma 4: $Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t))$ on $\mathbb{R}$ with $\sqrt{\Theta'(t)} > 0$. Hence $Z(t) = 0 \iff Y(\Theta(t)) = 0$ for $t \in \mathbb{R}$. By Theorem 8 and Axiom 7, $\{Y = 0\} = \{\widetilde{Y} = 0\} \subset \mathbb{R}$, so the zero set of Z on $\mathbb{R}$ equals $\Theta^{-1}(\{\widetilde{Y} = 0\}) \subset \Theta^{-1}(\mathbb{R}) = \mathbb{R}$. □

Theorem 10 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re s = \frac{1}{2}$.*

Proof. Let ρ be a nontrivial zero of ζ. By Axiom 6, $\rho = \frac{1}{2} + i\tau$ for some $\tau \in \mathbb{C}$ with $\zeta(\frac{1}{2} + i\tau) = 0$ and $0 < \Re(\frac{1}{2} + i\tau) < 1$, i.e. $|\Im\tau| < \frac{1}{2}$.

Write $\tau = \beta - i\alpha$ with $\beta, \alpha \in \mathbb{R}$ and $|\alpha| < \frac{1}{2}$. Then $\rho = \frac{1}{2} + \alpha + i\beta$, and $\Re\rho = \frac{1}{2} + \alpha$. RH is the statement $\alpha = 0$.

The pullback Y is the unitary stationary-frequency representation of Z (Definition 2); Theorem 8 supplies its entire extension $\widetilde{Y}$ of exponential type ≤ 1 with band $[-1, 1]$, real on $\mathbb{R}$. By Axiom 7, $\widetilde{Y}$ has no complex zeros: its zero set lies in $\mathbb{R}$.

By Lemma 4 and Theorem 9, the real-line relation

$$Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t)), \quad t \in \mathbb{R}, \qquad \sqrt{\Theta'} > 0,$$

together with the stationarity axiom, yields that the Hardy Z-function, considered as the warp-pullback $Z = \sqrt{\Theta'} \cdot Y \circ \Theta$ of the stationary representation, has no complex zeros either: any zero of Z obtained by pulling back a zero of $\widetilde{Y}$ through Θ sits at Θ^{-1} of a real point, hence at a real t.

By Axiom 6, the nontrivial zero $\rho = \frac{1}{2} + \alpha + i\beta$ corresponds to a Z-value at the parameter $t = \tau = \beta - i\alpha$, and $\zeta(\rho) = 0$ means this Z-value is zero. This Z-zero is the image through the warp pullback of a $\widetilde{Y}$-zero at $\Theta(\tau)$. Axiom 7 places that $\widetilde{Y}$-zero on $\mathbb{R}$. $\Theta : \mathbb{R} \to \mathbb{R}$ is a bijection (Lemma 4), so the preimage of a real point under Θ is real, giving $\tau \in \mathbb{R}$, i.e. $\alpha = 0$. Therefore $\Re\rho = \frac{1}{2}$. $\square$